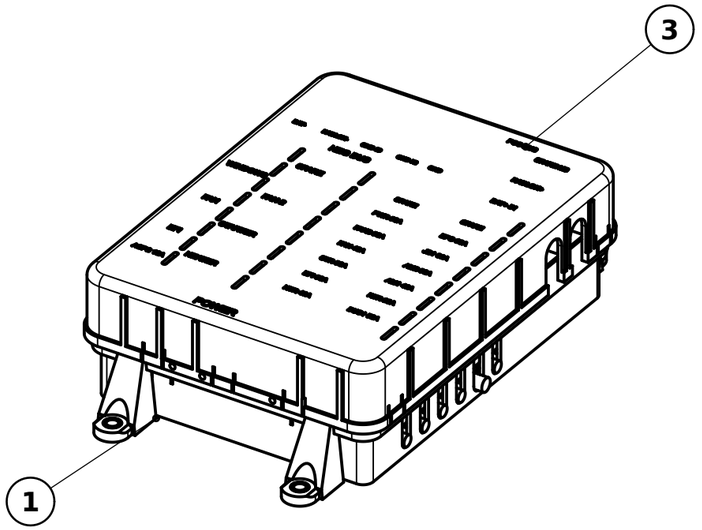
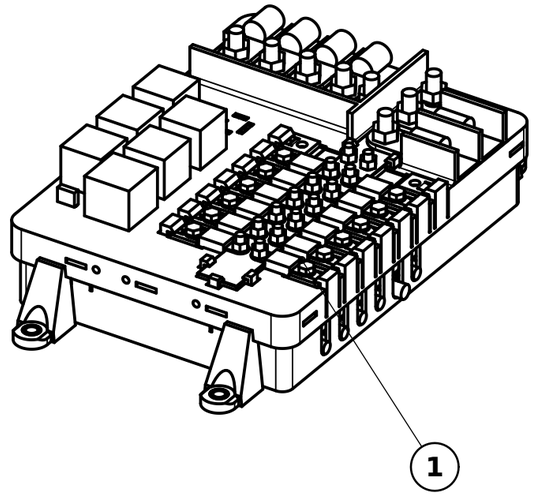
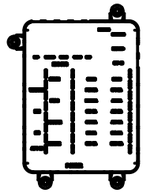
REVISIONS			
REV	DESCRIPTION	BY	DATE
U	#297: FB-101 base T, FB-103 cover G, FB-104 R1 clip D	Claude	18/10/26
V	#303: FB-101 base U, FB-102 pan G, FB-104 R1 clip E	Claude	18/10/26
W	Buildability review: strap hook 40 deg lead-in, strap seat pocket (R181) - FB-101 base V; first assembly drawing	Claude	18/10/26
X	FB-117 strap B (its PEM nuts in the strap model)	Claude	18/10/26
Y	FB-117 strap C, FB-101 base W, FB-102 pan H (N36 = A: lug PEM clear of the bend)	Claude	18/10/26
Z	N37: FB-117 strap D (FHS stud + DIN 934 top nut in a new printed holder, 14), FB-101 base X; lug bolts out, Keps nuts on the lug studs	Claude	19/10/26

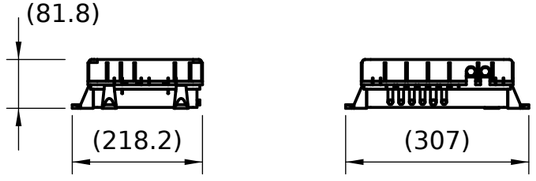


TRIMETRIC VIEW



TRIMETRIC - COVER REMOVED





FRONT / TOP / RIGHT VIEWS SCALE 1:10

ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL / SPEC
1	1	ST185-FB-101	POWER BASE (WITH THE BUS ZONE)	PA12, 3D PRINTED
2	1	ST185-FB-102	POWER PAN (BELLY PAN)	PA12, 3D PRINTED
3	1	ST185-FB-103	POWER COVER	PA12, 3D PRINTED
4	2	ST185-FB-113/-114	POS BUS BAR L226 + NEG BAR L105	C110 Cu, TINNED
5	1	ST185-FB-117	OUTPUT C-STRAP 15 x 3 (14 FROM 1 BAR)	C110 Cu, TINNED
6	14	PEM FHS-M5-12	FLUSH-HEAD SELF-CLINCHING STUD M5 X 12	C110 Cu, TINNED
7	14	RS 527-268	HEX NUT M5 DIN 934, TOP NUT IN ITS HOLDER	A2 STAINLESS
8	14	ST185-FB-118	C-STRAP TOP-NUT HOLDER, PA12 (1 PER STRAP)	PA12, 3D PRINTED
9	6	AMI-30	AMI BOLT-DOWN FUSE 30 A	PURCHASED
10	3	AMI-40	AMI BOLT-DOWN FUSE 40 A	PURCHASED
11	1	AMI-50	AMI BOLT-DOWN FUSE 50 A	PURCHASED
12	2	AMI-60	AMI BOLT-DOWN FUSE 60 A	PURCHASED
13	1	ATM-10	ACPS MINI BLADE FUSE 10 A	PURCHASED
14	1	1-1414687-0	HEATER RELAY K_HTR, 1 FORM C, 12 V, 680 OHM	PURCHASED
15	3	1-1414147-0	MAXI ISO F7 RELAY 680 OHM (k_efi/k_str/k_fan)	PURCHASED
16	2	1-1393304-0	RELAY K_FAN2 / POWER (SEE NOTE 2) (NOT SHOWN)	PURCHASED
17	6	280756-4	MAXI RELAY LOAD RECEPTACLE 9.5 X 1.2	PURCHASED
18	4	280755-4	MAXI RELAY LOAD RECEPTACLE 9.5 X 1.2, 10-7 AWG	PURCHASED
19	12	42281-1	RELAY COIL RECEPTACLE 6.3 X 0.8, 18-14 AWG	PURCHASED
20	2	62998-2	HEATER RELAY LOAD RECEPTACLE 6.3 X 0.8	PURCHASED
21	1	42281-1	HEATER RELAY 87A RECEPTACLE 6.3 X 0.8 (SEE NOTE 2) (NOT SHOWN)	PURCHASED
22	1	1-968851-1	MINI FUSE RECEPTACLE MCP 2.8, 16 / 14 AWG	PURCHASED
23	1	1-968849-1	MINI FUSE RECEPTACLE MCP 2.8, 20 / 18 AWG	PURCHASED
24	1	RNF-100-1/4-0-STK	INSULATION BUILD-UP SLEEVE 1/4, 2 ENDS X 15 MM (NOT SHOWN)	PURCHASED
25	1	RNF-100-1/8-0-STK	INSULATION BUILD-UP SLEEVE 1/8 (NOT SHOWN)	PURCHASED
26	1	RNF-100-3/32-0-STK	INSULATION BUILD-UP SLEEVE 3/32 (NOT SHOWN)	PURCHASED
27	1	ST185-FB-104	R1 RETAINER CLIP	PA12, 3D PRINTED
28	4	94459A140	M3 HEAT-SET INSERT (OPTIONAL COVER BACKUP) (NOT SHOWN)	BRASS
29	4	SSCF-M3-10-A4	M3 x 10 CAP SCREW (OPTIONAL COVER BACKUP) (NOT SHOWN)	A4 STAINLESS
30	9	Klauke 2R5	TUBULAR LUG 10 MM2, M5 HOLE (8-10 AWG)	Cu, TINNED
31	1	Klauke 3R5MS	TUBULAR LUG 16 MM2, M5 HOLE (6 AWG) (NOT SHOWN)	Cu, TINNED
32	2	36161	RING LUG 12-10 AWG, M5 / #10 HOLE (NOT SHOWN)	Cu, TINNED
33	6	Klauke 6R8	TUBULAR LUG 50 MM2, M8 HOLE	Cu, TINNED
34	14	RS 287-3921	HEX BOLT M5 x 10, EAR (DOWN INTO THE TOP NUT)	A2 STAINLESS
35	14	RS 263-2205	HEX BOLT M5 x 16 / M8 x 25, BAR STUDS	A2 STAINLESS
36	28	RS 264-6432	KEPS NUT M5 (BAR STUDS + LUG STUDS) / HEX NUT M8	A2 STAINLESS
37	8	RS 190-276	HEX BOLT M8 X 25 (BUS STUDS: 3 POS + 5 N (SEE NOTE 2) (NOT SHOWN)	A2 STAINLESS
38	8	RS 530-769	HEX NUT M8 (BUS STUDS) (SEE NOTE 2) (NOT SHOWN)	A2 STAINLESS
39	8	RS 797-6068	PLAIN WASHER M8 (BUS STUDS) (NOT SHOWN)	A2 STAINLESS
40	4	EFC 10S00056	COMPRESSION LIMITER, FEET (4)	STEEL, ZINC

NOTES:

1. SIZES IN BRACKETS ARE FOR REFERENCE. SHEETS 2 - 5: WHAT EACH FUSE AND RELAY FEEDS AND THE HARNESS WIRE AT EACH END.
2. PARTS THAT SHARE ONE MODEL (RELAYS, LUGS, NUTS, BOLTS) ARE BALLOONED ON THE FIRST ITEM OF THEIR GROUP.
3. M5 BOLTS AND NUTS: 4 N·m MAX (AMI FUSE SHEET).
4. C-STRAPS: BUILD THEM AS ST185-FB-117 SHEET 2, THEN PUSH EACH STRAIGHT DOWN INTO ITS WELL UNTIL THE HOOKS CLICK OVER THE LIPS.
5. TERMINALS GO UP FROM UNDER THE DECK UNTIL THE LANCE CLICKS; TUG THE WIRE TO CHECK (TE 114-2025 3.6, 114-20021 3.7).
6. TIE THE HARNESS NO CLOSER THAN 38 mm UNDER THE DECK; NO SHARP BENDS AT THE DECK (TE 114-20021 3.7 C).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	Z	SEE BOM

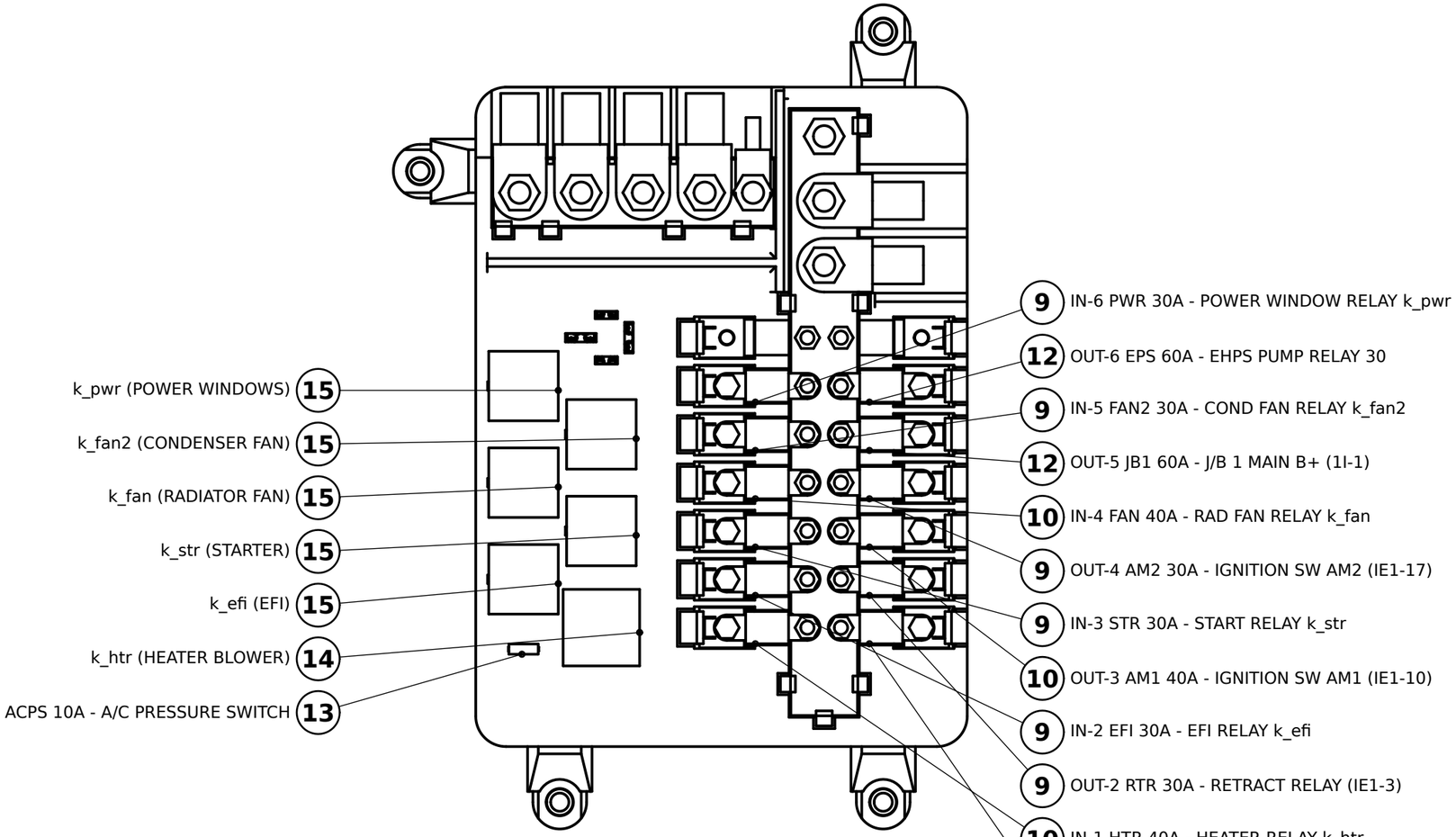
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

WHAT EACH FUSE AND RELAY FEEDS (COVER OFF)

TOP VIEW SCALE 1:2



UNLESS OTHERWISE SPECIFIED:
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TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
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COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	Z	SEE BOM

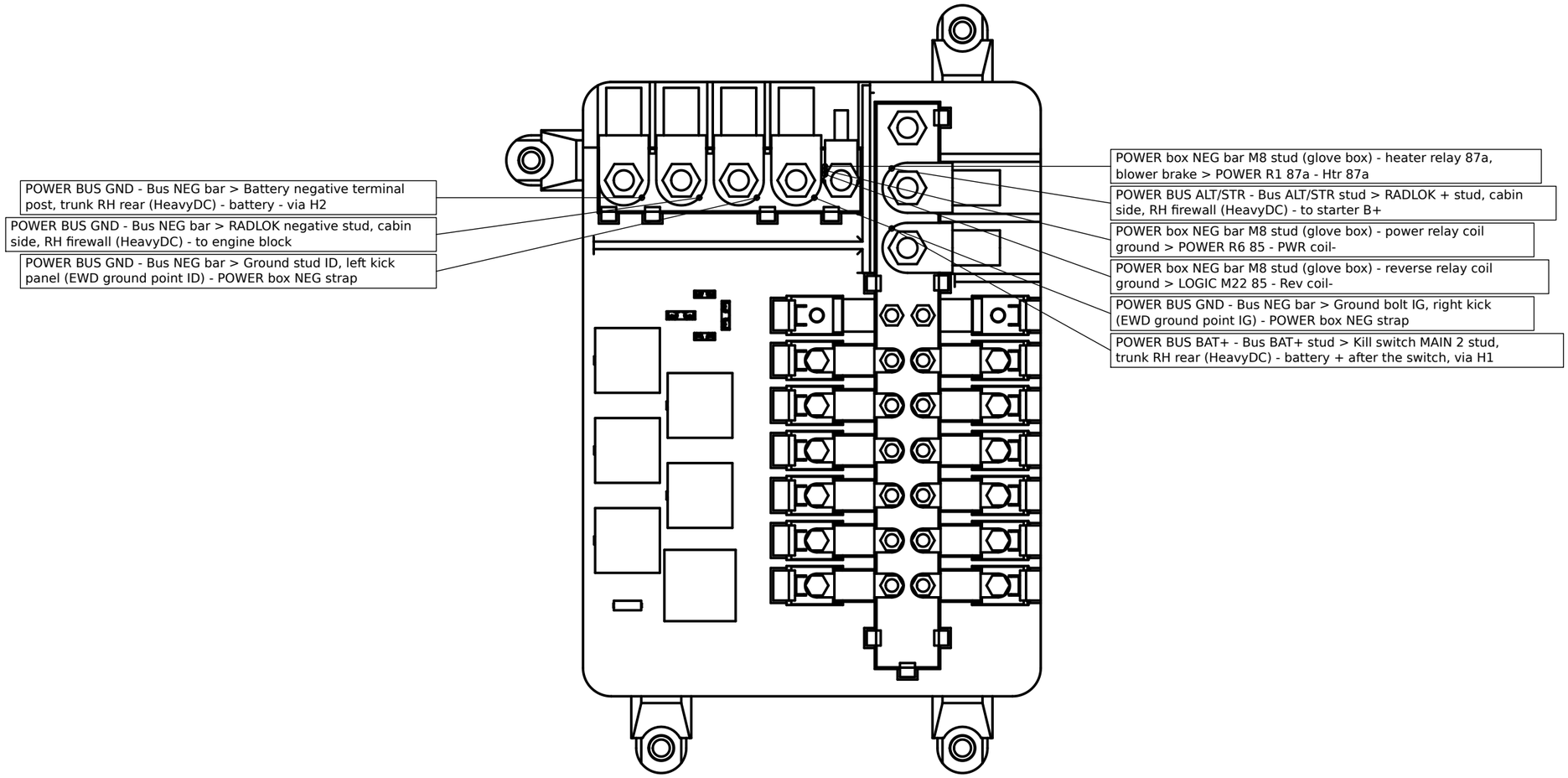
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

HARNESS WIRES AT THE BUS STUDS (COVER OFF)

TOP VIEW SCALE 1:2



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**Z**	SEE BOM

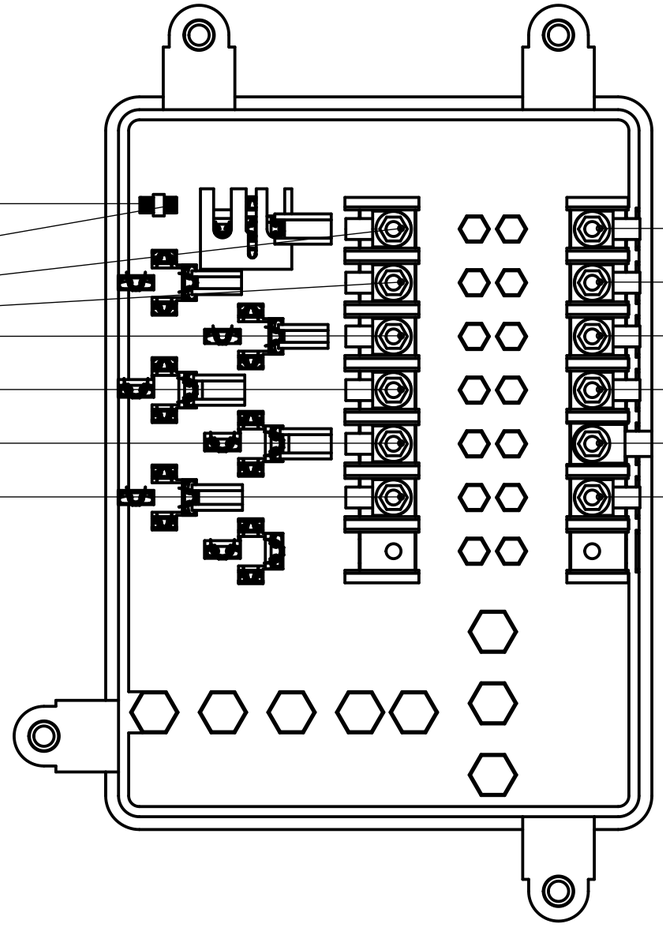
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

HARNESS WIRES AT THE C-STRAP LUGS AND THE ACPS FUSE - SEEN FROM BELOW, BELLY PAN OFF

BOTTOM VIEW SCALE 1:2



ACPS IN > Heater relay output splice (4 AWG) - blower motor + and the ACPS fuse

ACPS > A/C pressure switch A5 pin 1 (loom C, engine bay) - switch +12 V (1990 EWD p.152)

AMI IN-1 > POWER R1 87 - Htr 12V in

AMI IN-2 > POWER R2 87 - EFI 12V in

AMI IN-3 > POWER R3 87 - Start 12V in

AMI IN-4 > POWER R4 87 - Rad fan 12V in

AMI IN-5 > POWER R5 87 - Cond fan 12V in

AMI IN-6 > POWER R6 87 - PWR 12V in

AMI OUT-1 > FB-LOGIC BUS1 - Bus 1 in

AMI OUT-2 > OEM cowl plug IE1 pin 3 (LH kick), W wire - RTR 30A to retract relay R8 (EWD 1993 p.116)

AMI OUT-3 > OEM cowl plug IE1 pin 10 (LH kick), W wire - AM1 to ignition switch (EWD p.31)

AMI OUT-4 > OEM cowl plug IE1 pin 17 (LH kick), B-R wire - AM2 to ignition switch (EWD p.31)

AMI OUT-5 > OEM J/B 1 plug 1I pin 1 (LH kick), B wire - J/B 1 main B+ (EWD p.18-19)

AMI OUT-6 > EHPS RELAY 30 - EHPS 12V in

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**Z**	SEE BOM

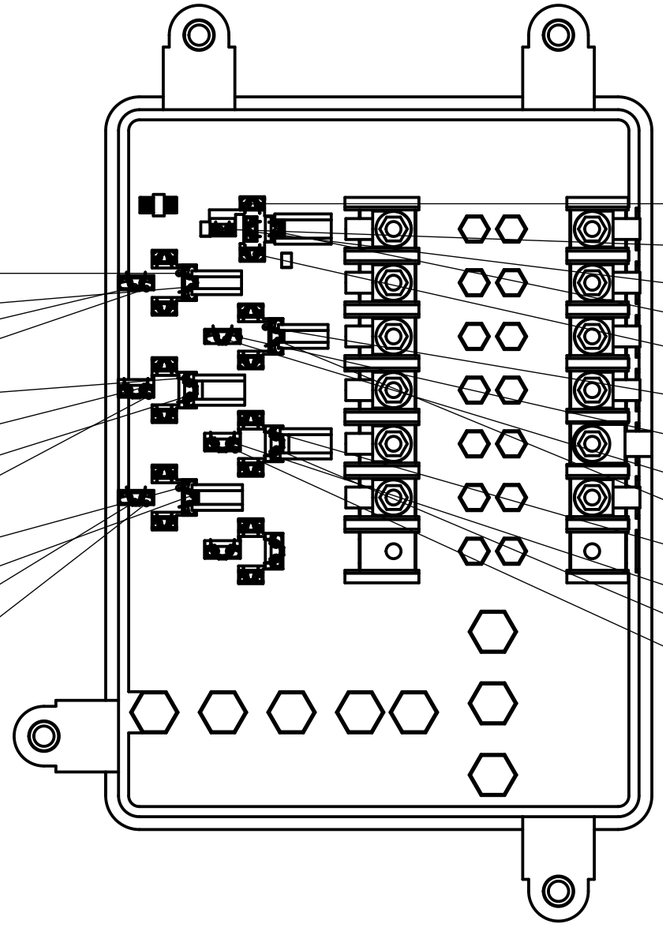
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SOURCE
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3RD ANGLE

HARNESS WIRES AT THE RELAYS - SEEN FROM BELOW, BELLY PAN AND R1 CLIP OFF

BOTTOM VIEW SCALE 1:2



k_efi 85 > ECU plug A pin 28 (Link ECU, center dash) - Aux 6 sinks the EFI relay coil

k_efi 87 > FB-POWER AMI IN-2 - EFI 30A

k_efi 30 > FB-LOGIC BUS2 - Bus 2 in

k_efi 86 > FB-LOGIC BUS1-1 - EFI86 5A

k_fan 85 > ECU plug A pin 29 (Link ECU, center dash) - Aux 5 sinks the radiator fan relay coil

k_fan 30 > Radiator fan plug pin + (loom C, front) - radiator fan +12 V, 8 AWG

k_fan 87 > FB-POWER AMI IN-4 - FAN 40A

k_fan 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 4 - SW12 +12 V bus

k_pwr 85 > POWER box NEG bar M8 stud (glove box) - power relay coil ground

k_pwr 87 > FB-POWER AMI IN-6 - PWR 30A

k_pwr 30 > OEM old R/B2 relay lead pin 1 (LH kick), W-G wire - power windows (EWD 1993 p.137)

k_pwr 86 > OEM old R/B2 relay lead pin 2 (LH kick), L-Y - GAUGE 15A, ignition on (EWD 1993 p.137)

k_htr 85 > OEM old R/B4 heater relay lead pin 1 (RH kick), L-W - A/C amp HR (EWD 1993 p.215)

k_htr 30 > Heater relay output splice (4 AWG) - blower motor + and the ACPS fuse

k_htr 87a > POWER box NEG bar M8 stud (glove box) - heater relay 87a, blower brake

k_htr 87 > FB-POWER AMI IN-1 - HTR 40A

k_htr 86 > OEM old R/B4 heater relay lead pin 3 (RH kick), R-L - coil + feed (EWD 1993 p.215)

k_str 85 > ECU plug A pin 26 (Link ECU, center dash) - Aux 8 sinks the start relay coil

k_str 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 6 - SW12 +12 V bus

k_str 30 > Starter solenoid S spade lug (A-engine loom) - start relay output

k_str 87 > FB-POWER AMI IN-3 - STR 30A

k_fan2 85 > ECU plug B pin 13 (Link ECU, center dash) - Ign 5 sinks the condenser fan relay coil

k_fan2 30 > Condenser fan plug pin + (loom C, front) - condenser fan +12 V, 8 AWG

k_fan2 87 > FB-POWER AMI IN-5 - FAN2 30A

k_fan2 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 7 - SW12 +12 V bus

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TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**Z**	SEE BOM

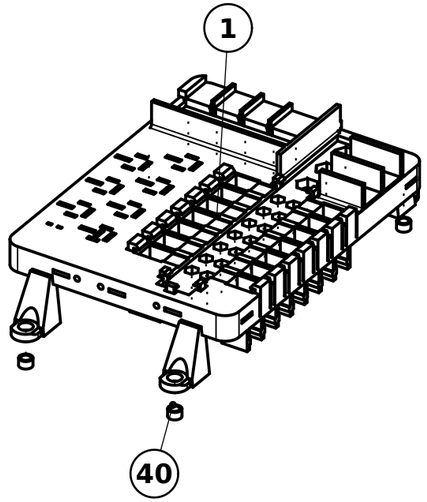
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3RD ANGLE

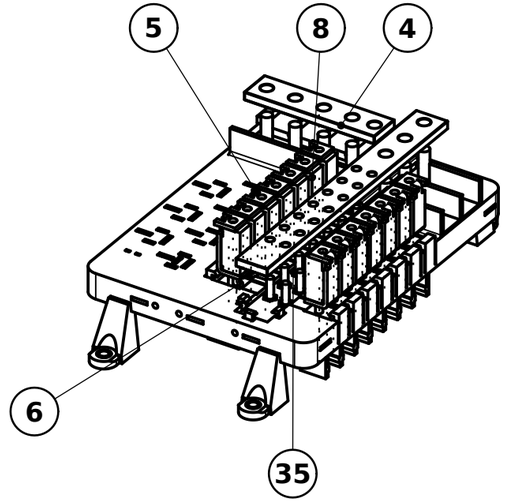
STEP 1: LIMITERS; BAR BOLTS FROM BELOW

EXPLODED TRIMETRIC (NOT TO SCALE)



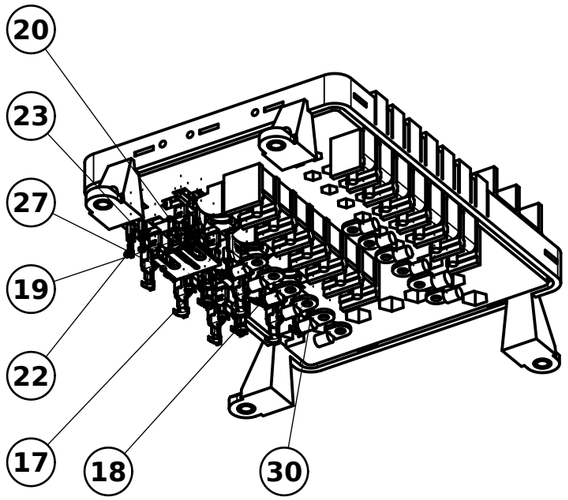
STEP 2: BARS; C-STRAPS STRAIGHT DOWN

EXPLODED TRIMETRIC (NOT TO SCALE)



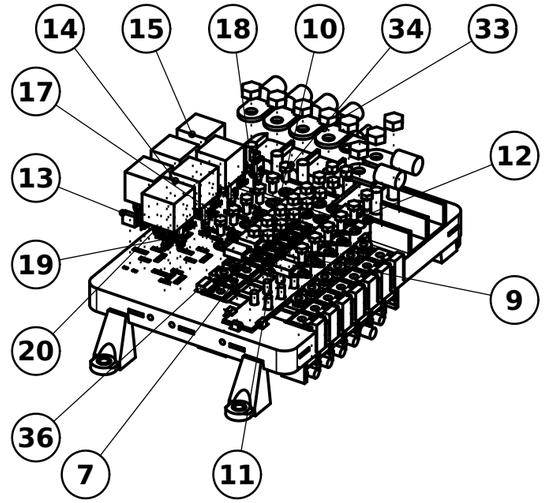
STEP 3: TERMINALS, R1 CLIP, LUGS UNDER THE STRAPS

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)



STEP 4: FUSES, EAR BOLTS, NUTS, LUGS, RELAYS

EXPLODED TRIMETRIC (NOT TO SCALE)



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DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**Z**	SEE BOM

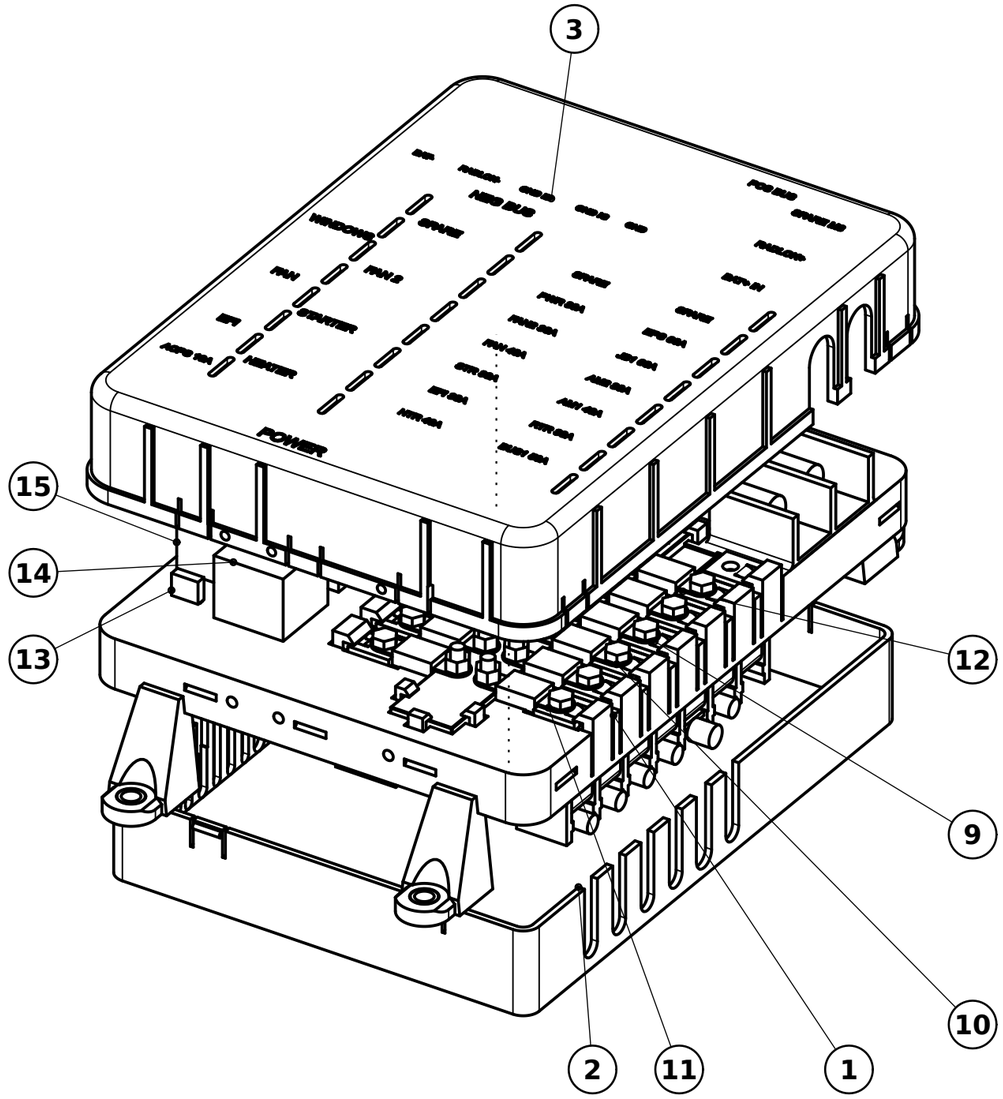
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

STEP 5: BELLY PAN + COVER

EXPLODED TRIMETRIC (NOT TO SCALE)



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**Z**	SEE BOM

SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
AS SHOWN	mm	B (11x17)	7 OF 7	2026-10-19	SteveCAD

SOURCE
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3RD ANGLE